

## 2.2 Problem solving and programming

**How computers can be used to solve problems and programs can be written to solve them**

***(Learners will benefit from being able to program in a procedure/imperative language and object oriented language.)***

### 2.2.1 Programming techniques

- (a) Programming constructs: sequence, iteration, branching.
- (b) Recursion, how it can be used and compares to an iterative approach.
- (c) Global and local variables.
- (d) Modularity, functions and procedures, parameter passing by value and by reference.
- (e) Use of an IDE to develop/debug a program.
- (f) Use of object oriented techniques.

### 2.2.2 Computational methods

- (a) Features that make a problem solvable by computational methods.
- (b) Problem recognition.
- (c) Problem decomposition.
- (d) Use of divide and conquer.
- (e) Use of abstraction.
- (f) Learners should apply their knowledge of:
  - backtracking
  - data mining
  - heuristics
  - performance modelling
  - pipelining
  - visualisation to solve problems.